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AIM 3410 ALUMINIUM PROFILE MACHINING CENTERS



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1. OVERVIEW

AIM 3410 is CNC controlled 4-axis machines designed specifically for machining all types of aluminium & PVC profiles, low carbon alloy and thin-walled steel profiles available on the market. With the 4th axis feature, it can also be machining on 3 surfaces of profiles.

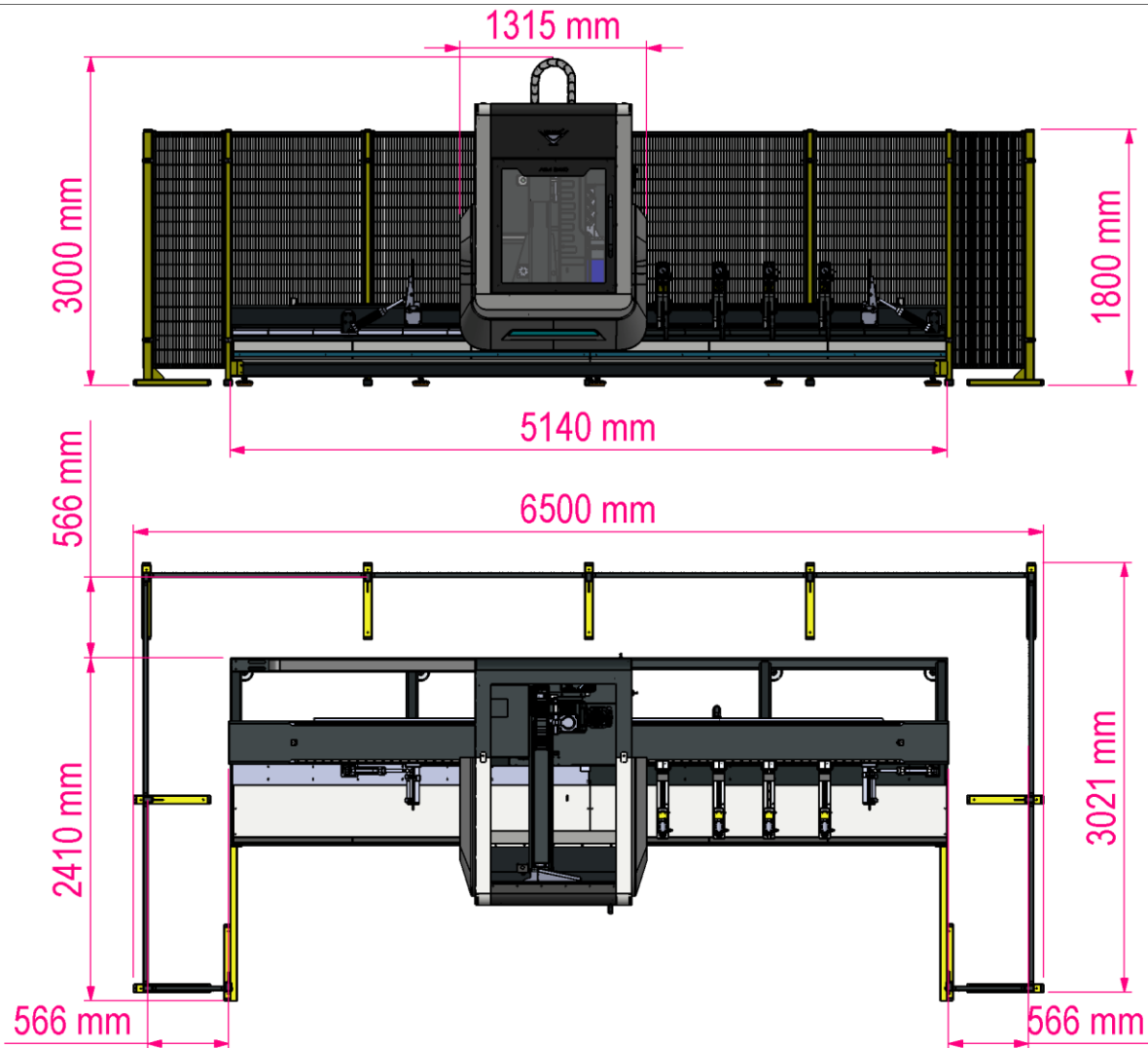
Quick tool changing is provided by the movable and in-line type magazine unit equipped with 8 standard tools. In order to minimize the tool change time, the tool magazine was designed on the moving Y axis of the Machine.

The machine provides a flexible processing opportunity especially in facade applications with its 180 mm diameter saw.

The machine has the ability to automatically position the profile clamps with CAMPROX program.

The front of the machine is equipped with a movable door with a transparent polycarbonate window to facilitate the monitoring of the workpiece processing by the operator.

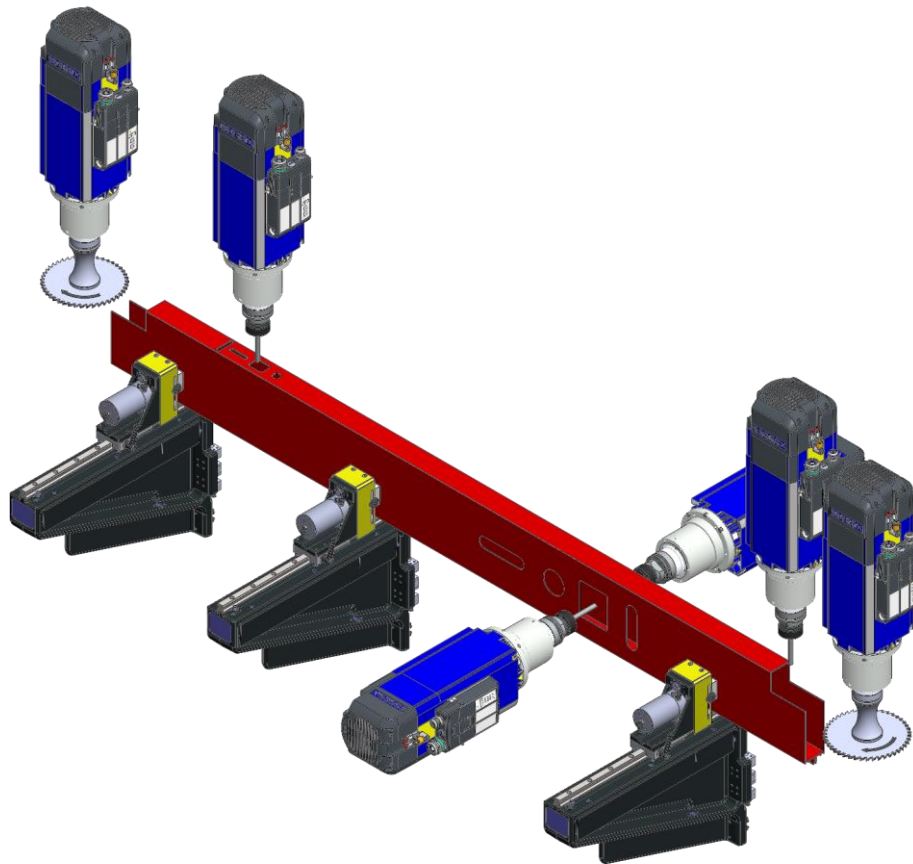
Thus, the safe operation of the operator is ensured.



2. TECHNICAL INFORMATION

2.1 General Features:

- Automation system providing CNC motion at 4-axis
- Tool magazine unite with the capacity of 7 pcs. of cutter tools and 1 pc. saw blade with 180 mm diameter
- Automatic clamp recognition and positioning feature
- PLC controlled spray tool cooling unit
- 4 pcs. of automatic clamps to catch the workpieces firmly.
- 1 x pneumatic rest used for resetting the workpiece in X direction (second is optional)
- Automatic central lubrication system for rails and bearings
- Cartridge lubrication system for X, Y and Z axis
- Special CAM program developed by Yilmaz engineering team
- Operator safety barrier
- 15" colour touch screen
- Program import via USB flash memory stick
- Standard ISO 'G' code programming language
- Technical support capability via remote connection by computer
- Multi-language support
- Special sound insulation cabinet around the machining center
- Solid steel construction of the body and dynamic performance gantry
- LED lighting in the cabinet
- Hand held controller allows you to prove the job before running at speed (optional)

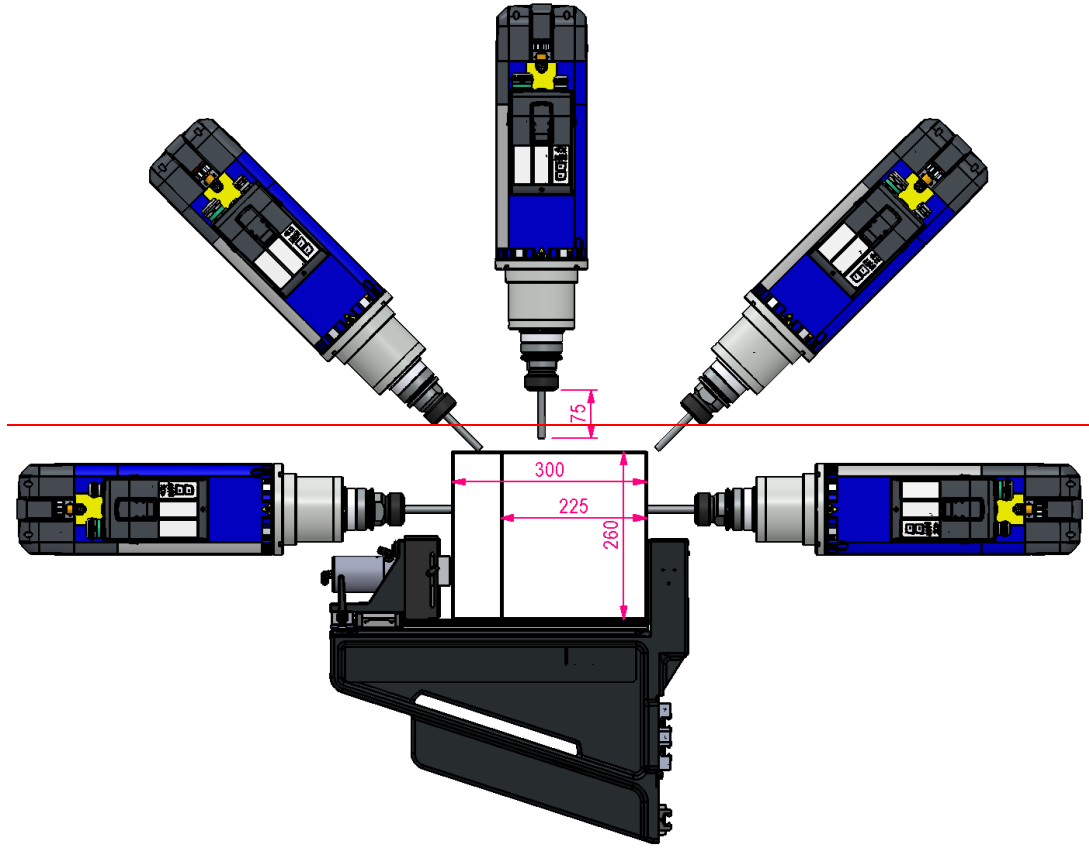


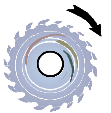
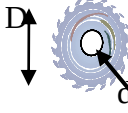
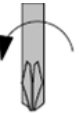
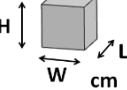
2-2 Technical Features

WORKING CAPACITY		AIM 3410
X-axis (mm)		3.200
Y-axis (mm) (top + back side surface)		*300
Z-axis (mm)		260
A-axis (°)		+95° / -95°
ELECTRO SPINDLE		
Max. speed (RPM)		24.000
Max. power S1 (kW)		7,5
Max. power S6 (kW)		9
Tool holder		HSK F63
AUTOMATIC TOOL MAGAZINE		
Tool capacity (7 pcs of cutter tools + saw blade)		8
Magazine type		Mobile with bridge
Max. saw dia. (mm)		180
Max. tool weight (kg)		3
FUNCTIONS		
Multi-piece operations (holes, slots, notching, saw blade cuts, etc.)		yes
MACHINABLE FACES		
With milling tool (top + side faces + end)		1+2+2
With saw tool (top + side faces + ends)		1+2+2
FEEDRATE		
Axis speed (m/min.)		X: 60, Y: 50, Z: 50, A: 72 °/sec,
Servo motors (kW)		X-axis:1, Y-axis: 1, Z-axis: 1, A-axis: 0.75
WORKPIECE CLAMPING		
Standard number of clamps		4
Max. optional number of the clamps		2
Vice positioning through the X-axis		automatic
GENERAL		
Net / gross weight (kg)		1900
Packing dimensions (W x L x H) (mm)		5140x2000x3000 /
Air supply required (bar)		6
Voltage-frequency		400V AC 3P PE 50-60 Hz,
Total power (kW)		11.6

YILMAZ reserve the rights to modify above specifications without notice

*In case of operation on top + back side + front side surfaces it is 225 mm

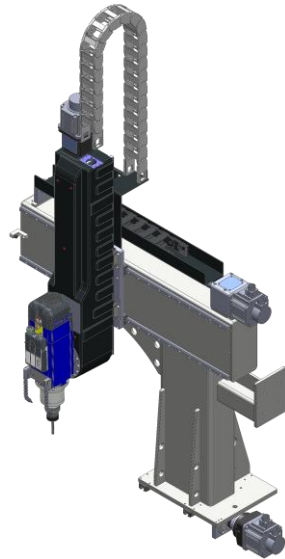


	TOPLAM GÜÇ						 kg	 kg
AIM 3410	11,6 kW	3000 RPM	D=180 mm d=30 mm	Ømax : 20 mm.	6 Bar 250 Lmin.	W=225 H=260 L=3200	1900	

3.FEATURES OF MACHINING CENTER

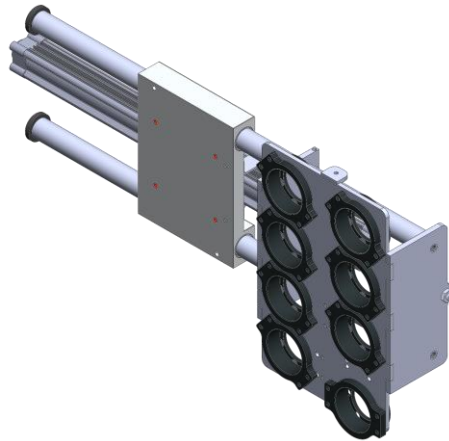
The parts of machining center are as following,

- Mono bloc machining center made of steel profile

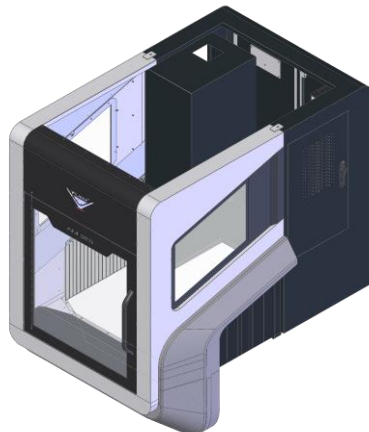


Milling Unit: 7,5 kW (max), 24.000 R.P.M HSD-motor
9 kW (max), 24.000 R.P.M HSD with encoder motor

- Magazine Unit with 8 pieces of tool jointing capacity



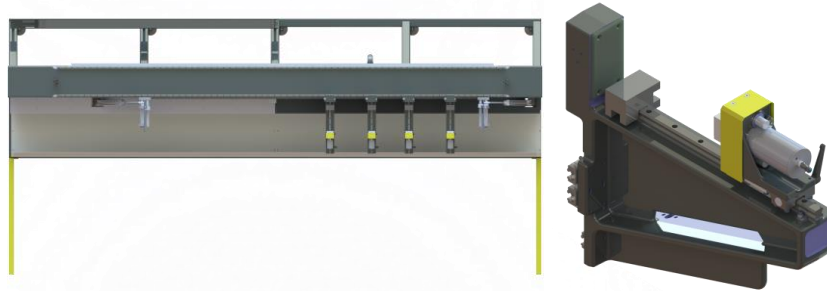
- Noise isolation cabinet around the machining head



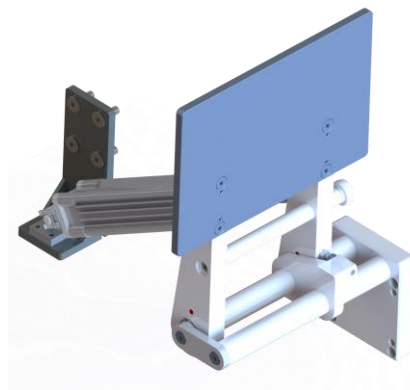
- Tool cooling unit with PLC controller



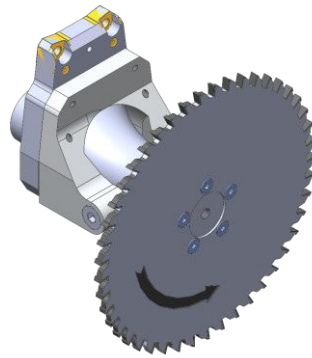
- 4 pieces of Automatically Operated Clamps to hold the workpiece (optionally maximum: 6)



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- 1 piece of pneumatically operated Support Units to reset the workpiece in X- direction



- Ø180 saw blade (optional)



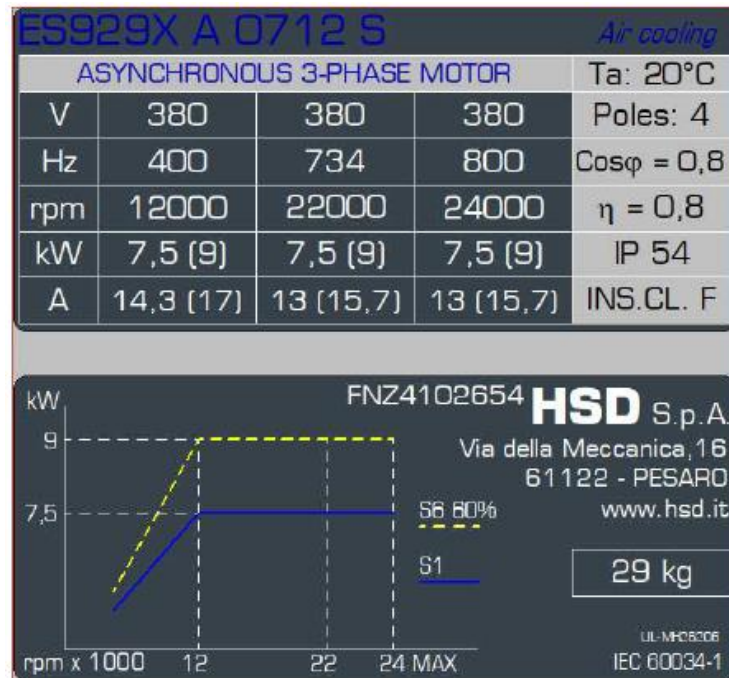
The entire machine is made of steel profiles and plates, and is designed in ergonomic dimensions that allow the operator to manually load the workpiece in a rapid manner.

It is painted with Electrostatic Powder Paint in order to ensure the corrosion resistance of metal parts.

The Front of the machine is equipped with a transparent polycarbonate window door to facilitate the operator to follow the machining of the workpiece. Thus, the safe operation of the operator has been ensured.

4. MAIN SPINDLE

The machining head is equipped with HSD brand milling unit with 6 poles, 7,5-9 kW power, 24,000 rpm rotation speed, and in HSK or F63 properties. The table below shows the performance of the milling unit depending on power and speed.



Tensione nominale (*)	Nennspannung (*)	Rated voltage (*)	V	380		380		380	
Frequenza nominale	Nennfrequenz	Rated frequency	Hz	400		734		800	
Velocità nominale	Nominale Geschwindigkeit	Rated speed	rpm	12000		22000		24000	
Tipo di servizio	Betriebsart	Duty type		S1 cont	S6 60%	S1 cont	S6 60%	S1 cont	S6 60%
Potenza nominale	Nennleistung	Rated power	kW	7.5	9	7.5	9	7.5	9
Coppia nominale	Nennmomento	Rated torque	Nm	6	7,1	3.2	3,9	3	3,6
Corrente nominale	Nennstrom	Rated current	A	14,3	17	13	15,7	13	15.7
Rendimento nominale η	Nennwirkungsgrad η	Rated efficiency η		0,8					
Fattore di potenza $\cos \varphi$	Leistungsfaktor $\cos \varphi$	Power factor $\cos \varphi$		0,8					
Numero di poli	Polzahl	Number of poles		4					
Classe di isolamento	Isolierklasse	Insulation class		F					
Tipo di raffreddamento	Kühlungstyp	Type of cooling		Elettroventola / Elektrolüfter / Cooling fan					
Peso	Gewicht	Weight	kg	29					

5. NUMERIC CONTROL

5.1 Numeric Control Hardware

Each numerical control hardware includes the below given components.

- Intel Atom E3815 CPU board
- 1460 MHz, single core
- 1 GB DDR3 SDRAM
- 512 kB L2 cache
- 1 piece USB3.0 ve 1 piece USB2.0 port
- 15" Colorful touch screen
- Speed control board for 4 axes and spindle

5.2 Numeric Control Software

Numerical Control Software provides the control of the following functions.

- Displaying position of axes on screen, clamps, rotational speed of spindle, manual control of axes, automatic cooling, profile support, machine safety cover, etc.
- Tool Table: It is used for storing tool parameters and can also be utilized for offset.
- CNC Setting Tables: Machine parameters which allows the maintenance (Axial length, origins, number of clamps, number of magazines, correction values)

6. AIMCAM PROGRAMMING

ISO programming language (G and M codes) is used for easy programming in the creation of macro programs. Thanks to the ISO programming language, it is very easy to write all kinds of machine operations, as well as the management of all machine functions can be easily provided. When the measurement values required by the macro-operation are entered by the operator, the machine automatically creates the program in the background and can perform the selected macro operation on the profile.

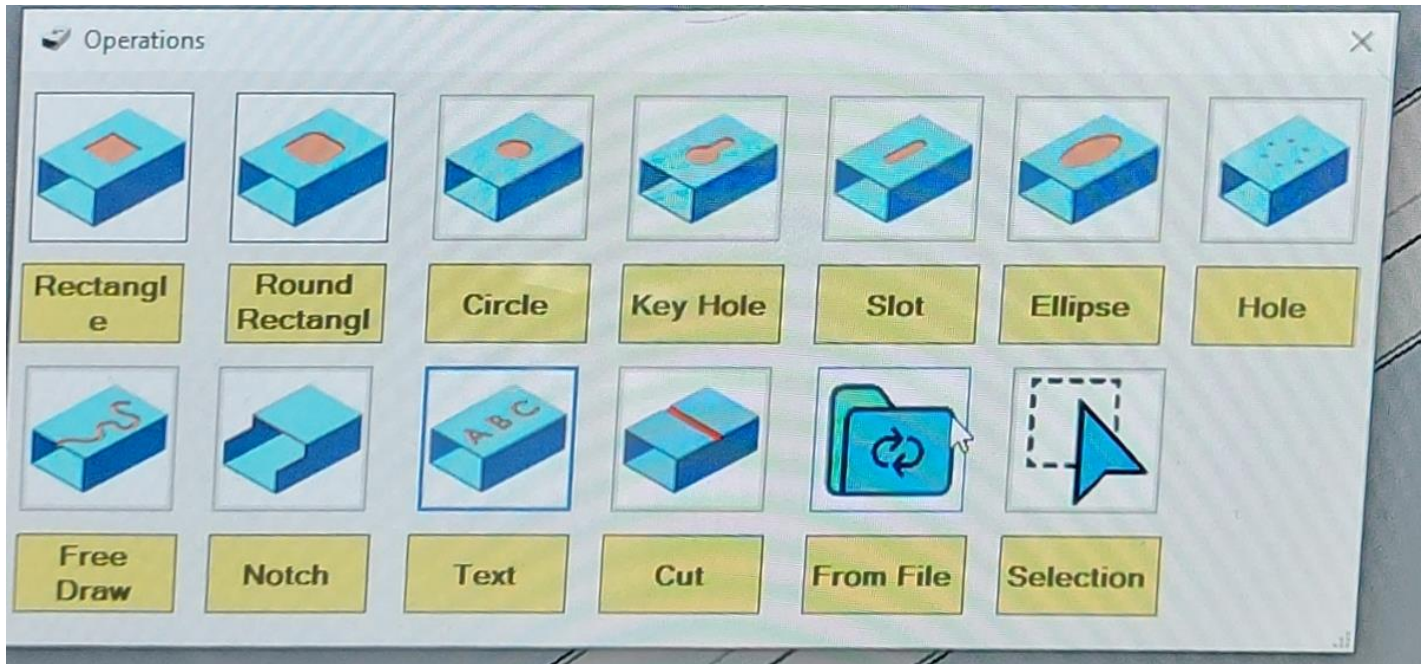
While programming the required macro, it is possible to change the speed, cutting speed, tool number, profile reference point in the same way.

In addition, the program can be run quickly by entering the profile height and width measurement from the screen, without the need to get a tool zero each time according to the profile.

With the AIMCAM program, customer-specific graphic drawings can be made and automatic G-codes can be created, and various operations can be easily performed on the profiles, and vice positioning can also be done automatically.

The AIMCAM program allows:

- Programming special milling operations,
- The ability to use saw,
- Automatic clamp handling before operation,
- Automatic clamp handling during operation,



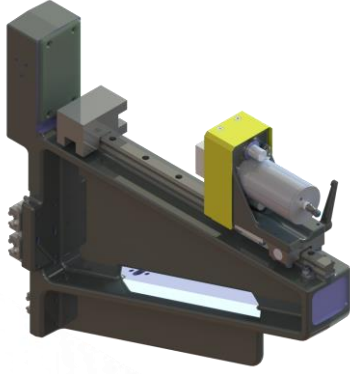
7.REMOTE CONNECTION SERVICE

It is possible to connect to the CNC machine remotely by Yilmaz Makine Technical Service. This connection is provided by a computer connected to the machine. This link provides fast program version changes, machine parameter changes or detailed analysis of machine parameters for maintenance purposes.

8.PROFILE CLAMPING SYSTEM

Profile clamping system is composed of 4-8 clamps. The clamps work as 2 and 4 by different hand buttons. The purpose of operating the clamps separately is to enable simultaneous machining of two separate profiles in the machining bed of the machine.

By means of CAMPROX software, the clamps are repositioned automatically by machine depending on the processes performed on workpiece.



9. DECLARATION OF CE CONFORMITY

Machine is delivered with CE sign and Certificate of CE Conformity to the customer. (Machine complies with the AT 2006/42EC, EN ISO 12100, 2006/95-EC Low Voltage Equipment, 2004/108-EC Electromagnetic Compatibility and EN 60204-1 standards).

9.1. Internal Tests

After the machine assembly is completed, the machine will be tested in accordance with the Internal Test Specification. These tests include mechanical checks, setting up different software parameters, machine tests, repeatability- accuracy tests, electrical and safety tests.

9.2. Training

Yilmaz Makine shall provide necessary training to customer.

The content of the training is as following;

- Introduction of Machining Center and its parts
- Knowledge of ISO programming language
- Explaining the basic information for the maintenance of the Machining Center
- To support the operator in making some customer-specific programs

For this training, the customer should assign an operator, who will be responsible of efficient operation and maintenance of machine.

10. DOCUMENTATION

Machine is delivered with the following documents to customer:

- AIM User and Maintenance Manuals
- Electrical and Pneumatic Circuit Schemes of the Machine
- AIMCAM Programming Manual
- SW Licenses

11. TERMS OF SALE

11.1. Guarantee Conditions

YILMAZ MAKİNE SANAYİ ve TİCARET A.Ş. guarantees that all its machines are tested before shipment and produced in accordance with international standards and reserves the right to make any changes to its products without prior notice.

General;

- The warranty terms apply only within the machine working time and warranty periods specified in the sales proposals.
- Warranty periods are based on the working period of 8 hours per day (1 shift).
- There may be "public holiday" delays in services and spare parts shipments.
- Yilmaz Makina is not responsible for the delays caused by Cargo in spare parts shipments.
- Installation and training of machines requiring installation must be done by authorized services or by Yilmaz machine technician.

Warranty coverage:

- All Factory (Yilmaz Machine) failures caused by manufacturing errors and defective parts replacements are made free of charge. Only shipping costs of the part (cargo, customs, etc.) are billed to the customer.
- If any fault or defective part of the machine is replaced in its place by our technical service personnel, the transportation, accommodation and operation costs of the service personnel belong to the customer.
- Even if the machine is covered by the warranty, service and parts shipping is not guaranteed in cases of user error. Therefore, the service costs are taken from the customer aside from transportation, accommodation and allowance costs.
- Even if the machine is covered by warranty, transportation, accommodation, allowance costs and service costs are charged to the customer during the relocation of the machine in the atelier or the transportation of the machine to another facility.

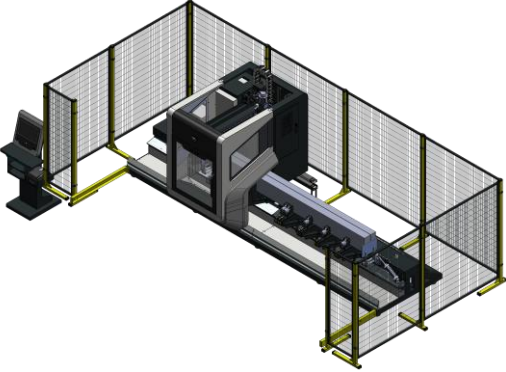
The warranty does not cover damages resulting from the following errors:

- Failure to comply with the rules set out in the user's manual,
- Malfunctions arising from incorrect electrical installations such as lack of phase due to incorrect voltage and voltage fluctuations, excess or low voltage
- Problems caused by lack of grounding in electrical installation,
- Failures as a result of not using air dryers in pneumatic products,
- Failures that are caused by working area ambient temperature outside +4°C to +40°C,
- Problems arising from failure to clean the machine,
- Failure to make the compulsory maintenance specified in the user's manual on time by Yilmaz Machinery or Authorized Services,
- Improper use or failure to use the machine for design purposes; machining outside the limits of the machine specified in the user's manual or technical document (profile sizes, tools, etc.),
- Use of machining tools and tool holders that are in poor quality and not suitable for the processed part,
- Electrical - electronic parts failures (caused by incorrect voltage),

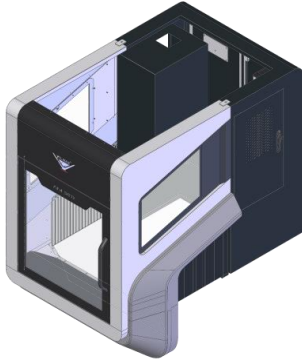
- Problems that may occur as a result of incorrect data input from the operator or incorrect data from optimization programs,
- Damage that may occur during relocation within the workshop or when the machine is moved to another facility,
- Natural disasters (lightning, fire, flood, etc.) or contact of the machine and / or the environment with rain or snow water
- Problems arising from failure to follow the instructions in the machine installation section in the user manual,
- Consumables worn in use (milling, saw, belt, machine coal, rollers, transparent protectors, protective plates) and products that need to be changed during periodical maintenance (bearing, belt, lubricant, rollers, etc.) are not covered by warranty.

11.2. Equipment and Service Included in Proposal

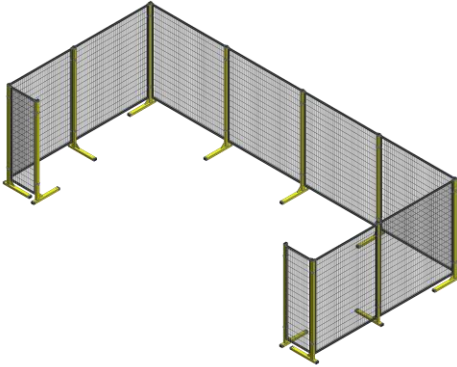
Equipment and services included in proposal are listed below:

Mechanical Parts	EURO
1 pc of 4-axis machining center having 3200 mm machining capacity 	
1 pc. operator safety barrier 	

1 pc of protection cabinet around machining head



1 set of safety fence around the machine

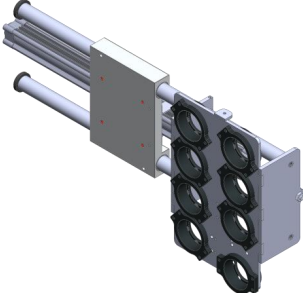


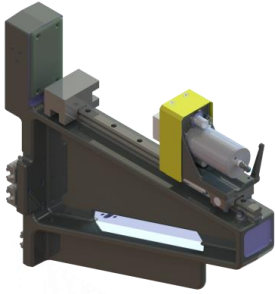
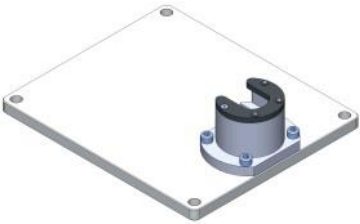
Software and Accessories

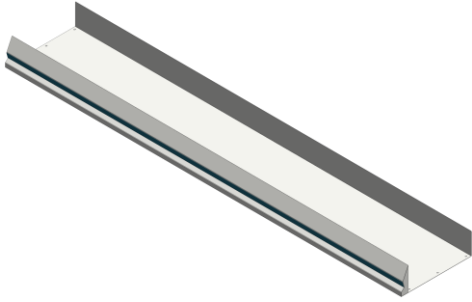
1 pc. of B&R operating system



1 pc of 8 tool capacity magazine unit

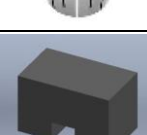
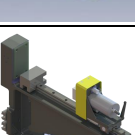


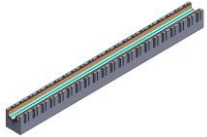
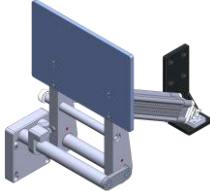
1 set (4-8 pcs) of profile holding clamps 	
11 pcs HSK F63 tool holding fork (assembled) 	
3 pieces of automatic tool cooling system 	
1 set of cartridge lubrication system 	
1 set of tool holder setup kit (assembled) 	

1 set of chip pan (assembled) 	

11.3 Optional equipment

AIM 4420 PROFIL PROCESSING CENTER OPTIONAL PART LIST			
Sıra no (item no)	Parça resmi (picture)	Parça kodu (part number)	Parça adı (part name)
1		3UA920030-0050	AIM 7510 TOOL & APARATUS KIT - 6 pc cutters (2 pc. Ø5, 1 pc. Ø6, 1 pc. Ø8, 1 pc. Ø10, 1 pc. Ø12) -12 pc. HSK F63 Tool holders -6 pc. Collets (3 pc. Ø5-6, 1 pc. Ø7-8, 1 pc. Ø9-10, 1 pc. Ø11-12) -12 pc. mould height
2		1SK050000-0038	ÇAP Ø 5*15 L50-Z1 KARBUR FLEX-TEK AĞIZLI FREZE Ø 5*15 L50-Z1 CARBIDE CUTTER
3		1SK050000-0039	ÇAP Ø 6*18 L57-Z1 KARBUR FLEX-TEK AĞIZLI FREZE Ø 6*18 L57-Z1 CARBIDE CUTTER
4		1SK050000-0040	ÇAP Ø 8*20 L63-Z1 KARBUR FLEX-TEK AĞIZLI FREZE Ø 8*20 L63-Z1 CARBIDE CUTTER
5		1SK050000-0018	ÇAP Ø 10*72-Z2 KARBUR FLEX- ÇİFT AĞIZLI FREZE Ø 10*72-Z2 CARBIDE CUTTER
6		1SK050000-0072	ÇAP Ø 12*83-Z2 KARBUR FLEX-ÇİFT AĞIZLI FREZE Ø 12*83-Z2 CARBIDE CUTTER
7		1SK050000-0056	ÇAP Ø13-Z2 KARBUR FREZE- ÇİFT AĞIZLI FREZE Ø13-Z2 CARBIDE CUTTER

8		3UB410030-0018	AIM 3410 TESTERE GRUBU
9		1SK050000-0256	TAKIM TUTUCU HSK F63 TOOL HOLDER
10		1SK110000-0401	PENS 4 LUK ISO 30 (ER32) PENS ER 32 COLLET (Ø3-4)
11		1SK050000-0023	470 E-6 ER32 Q6 PENS (Ø5-6) 470 E-6 ER32 Q6 COLLET (Ø5-6)
12		1SK050000-0057	470 E-8 ER32 Q8 PENS (Ø7-8) 470 E-8 ER32 Q8 COLLET (Ø7-8)
13		1SK050000-0024	470 E-10 ER32 Q10 PENS (Ø9-10) 470 E-10 ER32 Q10 COLLET (Ø9-10)
14		1SK050000-0073	470 E-12 ER32 Q12 PENS (Ø11-12) 470 E-12 ER32 Q12 COLLET (Ø11-12)
15		1SK050000-0058	470 E-14 ER32 Ø14 PENS (Ø13-14) 470 E-14 ER32 Ø14 COLLET (Ø13-14)
16		1SK050000-0048	ER32-Ø20 PENS (Ø19-20) ER32-Ø20 COLLET (Ø19-20)
17		2TU013210-0196	PROFİL YÜKSELTME TAKOZU MOULD HEIGHT PART
18		3UB410030-0011	AIM 3410 MENGENE GRUBU AIM 3410 CLAMPING UNIT
19		3UB410030-0016	AIM 3410 TAKIM BOYU ÖLÇME GRUBU AIM 3410 TOOL LENGTH MEASUREMENT UNIT

20		1SK010000-0114	Ø180*5*30 Z=45 ELMAS UÇLU TESTERE (AIM 4310)
21			ÖZEL PROFİL APARAT KALIBI/ SPECIAL MOLD HEIGHT DESING
22		1HD030000-0068	GRES PPOMPASI- MG 70 GREASE GUN UNIT-MG 70
23	AIMCAM SW	0AA020000-0848	YILMAZCAM 4 AXIS YAZILIM PROGRAMI (AIM 3410)
24		3UB410030-0006	AIM 3410 DAYAMA GRUBU SAĞ

11.4 Equipment and Services Excluded from Proposal

The below equipment and services are excluded in price proposal.

- Customer is responsible for electrical and pneumatic connections of machine before delivery
- Special Apparatuses
- Customer specific machining programs
- Shipment